

PHYS1408-002 Final Exam

Wednesday December 9, 2015 7:30-10pm

$$\rho_{\text{air}} = 1.21 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{ice}} = 917 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{water}} = 1000 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{seawater}} = 1028 \frac{\text{kg}}{\text{m}^3}$$

$$P_0 = 1.013 \times 10^5 \frac{\text{N}}{\text{m}^2} \quad G = 6.67 \times 10^{-11} \frac{\text{N m}^2}{\text{kg}^2} \quad c = 3 \times 10^8 \text{ m/s}$$

$$M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg} \quad R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$$

1. The scientific method requires that you construct something based on your observations and test it. What must be constructed and tested?

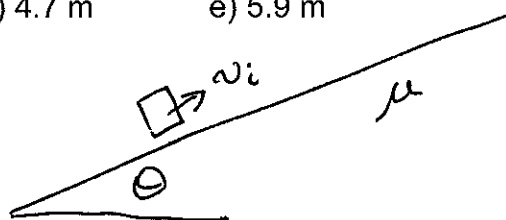
- a) table top model b) alibi c) explanation d) mental model e) procedure

2. Aunt Matilda goes to a well and throws a penny straight down the well at 2 m/s. She hears a splash after 0.8 seconds. How deep is the well?

- a) 1.5 m b) 2.1 m c) 3.6 m d) 4.7 m e) 5.9 m

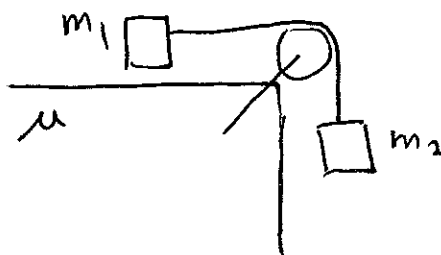
3. A block with mass 20 kg is sliding up a plane (coefficient of friction 0.3, inclined at 20°) at a speed of 3 m/s to the right (positive x-direction). How far does it go up along the plane before it comes to rest momentarily?

- a) 0.74 m b) 1.1 m c) 2.6 m d) 4.7 m e) 12 m



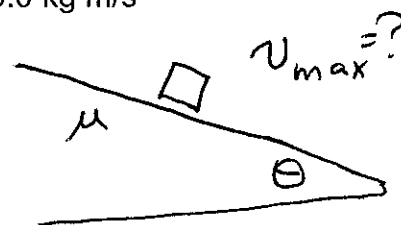
4. A mass (5 kg) rests on a surface ($\mu_{\text{static}}=0.6$) connected by a cord to a hanging mass of 2 kg. The connecting cord is mass-less and stretch-free. The pulley is mass-less and frictionless. What is the **force of friction** on the heavier mass?

- a) 9.8 kg m/s² b) 19.6 kg m/s² c) 29.4 kg m/s² d) 39.2 kg m/s² e) 49.0 kg m/s²



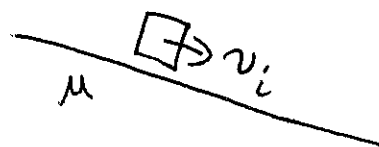
5. A car of mass 1000 kg turns a 10° inclined corner of radius 80 m that has a coefficient of friction of 0.2. What is the maximum speed the car can have around this corner without sliding up the incline?

- a) 17 m/s b) 18 m/s c) 19 m/s d) 20 m/s e) 21 m/s

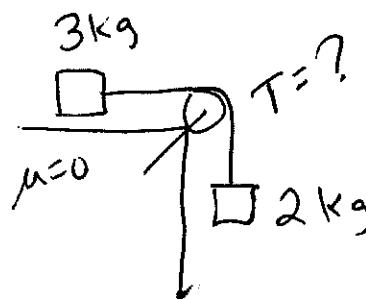


6. A block of mass 2 kg is sliding down an incline that has a coefficient of kinetic friction of 0.4. The block slows from 3 m/s initially to rest. The incline is at 20° . What was the magnitude of the force of friction during the time the block slowed?

- a) 1.8 kg m/s² b) 2.9 kg m/s² c) 4.8 kg m/s² d) 6.1 kg m/s² e) 7.4 kg m/s²

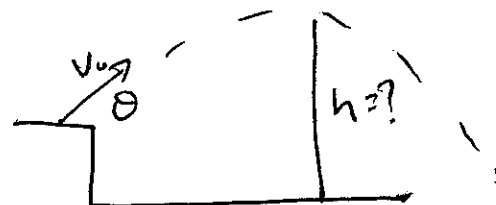


7. Mass 3 kg is resting on the table. Mass 2 kg is hanging off the side of the table as shown. They are connected by a cord over a pulley. The surface is frictionless. What is the tension in the string when the masses are released?
- a) 8 N b) 12 N c) 20 N d) 29 N e) 49 N



8. A wooden block (mass 0.01 kg) is set on a rotating record. The mass is 0.15 m from the axis of rotation and is moving with the disk at 0.2 m/s. What minimum coefficient of friction is required to ensure the mass does not slide outward?
- a) 0.02 b) 0.04 c) 0.1 d) 0.2 e) 0.4

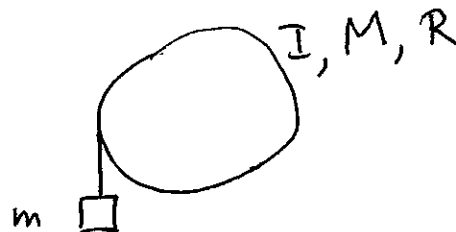
9. A mass is fired from a height of 2 m above the flat ground below (as shown) at 10 m/s at an angle of 30° above the horizontal. How high above the ground below does the mass fly at the highest point in its trajectory?



- a) 0.2 m b) 1.2 m c) 1.9 m d) 3.3 m e) 8.1 m
10. In order to go all the way around a frictionless circular loop of radius $R=3\text{m}$, how fast must a mass be moving as it enters the bottom of the loop?
- a) 7 m/s b) 9 m/s c) 11 m/s d) 13 m/s e) 15 m/s
11. A mass of 5 kg is sliding up a ramp ($\mu=0.5$ and $\theta=15^\circ$), initially sliding at 3 m/s. How long does it continue to slide along the ramp before it comes to rest?
- a) 0.4 s b) 0.7 s c) 1.6 s d) 2.1 s e) 4.7 s
12. A railroad car of mass m and velocity v collides completely **inelastically** with another railroad car of mass $0.5m$ and velocity $0.5v$. The two cars collide and couple together. What fraction of the initial kinetic energy is still retained by the cars after the collision?
- a) <1% b) 2-5% c) 10-20%
d) 30-50% e) >90%

13. A bullet (mass 0.005 kg) is fired into a suspended block of wood (5 kg). Initially the wood is not moving. The bullet lodges in the wood. Recoil of the wood causes it to swing (rising 0.01 m). What fraction of the initial kinetic energy remained in the motion of the block of wood? FYI- this is very close to a real situation (i.e., realistic numbers!)
- a) 0.1% b) 0.5% c) 1% d) 10% e) >20%

14. A mass of 0.05 kg hangs from a pulley ($I = \frac{3}{5}MR^2$, $M=3\text{ kg}$, $R=0.1\text{ m}$) as shown. What is the downward linear acceleration of the hanging mass?



- a) 8.5 m/s^2 b) 3.0 m/s^2 c) 1.1 m/s^2 d) 0.51 m/s^2 e) 0.26 m/s^2

15. A sphere ($M=0.2\text{ kg}$, $R=0.01\text{ m}$, $I = \frac{2}{5}MR^2$) rolls down an incline from a height of 0.5 m. At the bottom of the incline, what is the sphere's rotational velocity, ω ?
- a) 1.5 rad/s b) 26 rad/s c) 31 rad/s d) 270 rad/s e) 910 rad/s

16. A disk $I = \frac{1}{2}MR^2$, mass 200 kg, radius 3 m rotates at 3 rad/s. A wheel ($I = 0.8MR^2$ mass 100 kg, radius 2.5 m) initially rotating in the opposite direction at -3 rad/s is dropped onto the disk. Their collision is completely inelastic. After they stop bouncing what is the angular velocity (rad/s) of the two-object stack?

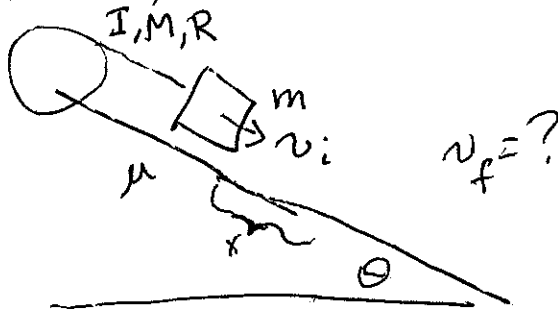
- a) -1.6 rad/s b) -0.86 rad/s c) 0 rad/s d) +0.86 rad/s e) +1.6 rad/s

17. A merry go round (a disk $I = \frac{1}{2}MR^2$, $M=500$ kg, $R=2$ m) rotates at +0.1 rad/s. A child with a mass of 50 kg jumps onto the merry go round (going same direction) at the rim. The child was running at 10 m/s as she jumps on. What is the new rotation rate for the merry go round WITH the child?

- a) 0.15 rad/s b) 0.2 rad/s c) 0.4 rad/s d) 0.7 rad/s e) 0.9 rad/s

18. A mass of 5 kg slides down a 20° inclined surface ($\mu=0.1$) at 0.80 m/s. It is connected to a pulley (as shown). The pulley has mass 2 kg, radius 0.05 m, and moment of inertia $0.5 MR^2$. How fast is the mass sliding when it has slid 0.5 m down along the incline?

- a) 2.1 m/s b) 1.6 m/s c) 1.1 m/s d) 0.9 m/s
e) it stops before it slides the whole 0.5 m



19. A 150 kg man (and the load he carries) is going up a leaning 20 kg ladder. The ladder is set at 70° . The floor and ladder have a coefficient of friction of 0.2. The wall is frictionless. How far up the 5 m long ladder can he safely climb?

- a) 5 m b) 4 m c) 3 m d) 2 m e) 1 m

20. A load of 400 kg is hung from a circular wire 4.00 m long, diameter 0.0005 m, and a Young's modulus of 8.00×10^{10} N/m². By how much does the wire stretch?

- a) 0.0001 m b) 0.001 m c) 0.01 m d) 0.1 m e) 1. m

21. A rocket fired from earth's surface rises 2000 km. At what speed was it launched?

- a) 600 m/s b) 3.1 km/s c) 4.0 km/s d) 5.5 km/s e) 11 km/s

22. What is the period of the circular orbit of two equal mass stars (10^{32} kg each) around their common center of mass? Separation distance $d=10^{11}$ m

- a) 57 minutes b) 12 hours c) 17 hours d) 20 days e) 190 days

23. What is the acceleration due to gravity on a 90 kg man at a height of 700 km above the Earth's surface?

- a) 5 m/s² b) 6 m/s² c) 7 m/s² d) 8 m/s² e) 9 m/s²

24. The earth and a moon (mass 10^{23} kg) are separated by 500,000 km, from center to center. How far from the center of the earth toward the moon would you have move to find a place where the gravitational pull towards the earth would equal the gravitational pull towards the moon?

- a) 50% b) 88% c) 95% d) 99% e) 99.99%

25. A spaceship passes you at $v=0.5c$. How long passes for you while 1 year passes in the ship?

- a) a year and twenty five seconds
b) a year and five minutes
c) a year and two hours
d) a year and five days
e) a year and two months